HARD · ADVANCED MATH · ANSWER KEY

Nonlinear functions

03

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ANSWER KEY · SAT QUESTION BANK

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Q1

B

B. 6

Choice B is correct. It's given that a quadratic function models the projectile's height, in meters, above the ground in terms of the time, in seconds, after it was launched. It follows that an equation representing the model can be written in the form $fx = ax - h^2 + k$, where fx is the projectile's estimated height above the ground, in meters, x seconds after the launch, a is a constant, and k is the maximum height above the ground, in meters, the model estimates the projectile reached h seconds after the launch. It's given that the model estimates the projectile reached a maximum height of 51.1 meters above the ground 3 seconds after the launch. Therefore, $k = 51.1$ and $h = 3$. Substituting 51.1 for k and 3 for h in the equation $fx = ax - h^2 + k$ yields $fx = ax - 3^2 + 51.1$. It's also given that the model estimates that the projectile was launched from an initial height of 7 meters above the ground. Therefore, when $x = 0$, $fx = 7$. Substituting 0 for x and 7 for fx in the equation $fx = ax - 3^2 + 51.1$ yields $7 = a0 - 3^2 + 51.1$, or $7 = 9a + 51.1$. Subtracting 51.1 from both sides of this equation yields $-44.1 = 9a$. Dividing both sides of this equation by 9 yields $-4.9 = a$. Substituting -4.9 for a in the equation $fx = ax - 3^2 + 51.1$ yields $fx = -4.9x - 3^2 + 51.1$. Therefore, the equation $fx = -4.9x - 3^2 + 51.1$ models the projectile's height, in meters, above the ground x seconds after it was launched. The number of seconds after the launch that the model estimates that the projectile will return to a height of 7 meters is the value of x when $fx = 7$. Substituting 7 for fx in $fx = -4.9x - 3^2 + 51.1$ yields $7 = -4.9x - 3^2 + 51.1$. Subtracting 51.1 from both sides of this equation yields $-44.1 = -4.9x - 3^2$. Dividing both sides of this equation by -4.9 yields $9 = x - 3^2$. Taking the square root of both sides of this equation yields two equations: $3 = x - 3$ and $-3 = x - 3$. Adding 3 to both sides of the equation $3 = x - 3$ yields $6 = x$. Adding 3 to both sides of the equation $-3 = x - 3$ yields $0 = x$. Since 0 seconds after the launch represents the time at which the projectile was launched, 6 must be the number of seconds the model estimates that the projectile will return to a height of 7 meters.

Alternate approach: It's given that a quadratic function models the projectile's height, in meters, above the ground in terms of the time, in seconds, after it was launched. It's also given that the model estimates that the projectile was launched from an initial height of 7 meters above the ground and reached a maximum height of 51.1 meters above the ground 3 seconds after the launch. Since the model is quadratic, and quadratic functions are symmetric, the model estimates that for any given height less than the maximum height, the time the projectile takes to travel from the given height to the maximum height is the same as the time the projectile takes to travel from the maximum height back to the given height. Thus, since the model estimates the projectile took 3 seconds to travel from 7 meters above the ground to its maximum height of 51.1 meters above the ground, the model also estimates the projectile will take 3 more seconds to travel from its maximum height of 51.1 meters above the ground back to 7 meters above the ground. Thus, the model estimates that the projectile will return to a height of 7 meters 3 seconds after it reaches its maximum height, which is 6 seconds after the launch.

Choice A is incorrect. The model estimates that 3 seconds after the launch, the projectile reached a height of 51.1 meters, not 7 meters.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q2

A

A. 8

Choice A is correct. It's given that the function P models the population, in thousands, of a certain city t years after 2003. The value of the base of the given exponential function, 1.04, corresponds to an increase of 4% for every increase of 1 in the exponent, $\frac{6}{4}t$. If the exponent is equal to 0, then $\frac{6}{4}t = 0$. Multiplying both sides of this equation by $\frac{4}{6}$ yields $t = 0$. If the exponent is equal to 1, then $\frac{6}{4}t = 1$. Multiplying both sides of this equation by $\frac{4}{6}$ yields $t = \frac{4}{6}$, or $t = \frac{2}{3}$. Therefore, the population is predicted to increase by 4% every $\frac{2}{3}$ of a year. It's given that the population is predicted to increase by 4% every n months. Since there are 12 months in a year, $\frac{2}{3}$ of a year is equivalent to $\frac{2}{3}12$, or 8, months. Therefore, the value of n is 8.

Choice B is incorrect. This is the number of months in which the population is predicted to increase by 4% according to the model $Pt = 2601.04^{t}$, not $Pt = 2601.04^{\frac{6}{4}t}$.

Choice C is incorrect. This is the number of months in which the population is predicted to increase by 4% according to the model $Pt = 2601.04^{\frac{4}{6}t}$, not $Pt = 2601.04^{\frac{6}{4}t}$.

Choice D is incorrect. This is the number of months in which the population is predicted to increase by 4% according to the model $Pt = 2601.04^{\frac{1}{6}t}$, not $Pt = 2601.04^{\frac{6}{4}t}$.

Q3

168

The correct answer is 168. The quadratic function g gives the estimated depth of the seal, gt, in meters, t minutes after the seal enters the water. It's given that function g estimates that the seal reached its maximum depth of 302.4 meters 6 minutes after it entered the water. Therefore, function g can be expressed in vertex form as $gt = at - 6^2 + 302.4$, where a is a constant. Since it's also given that the seal reached the surface of the water after 12 minutes, g12 = 0. Substituting 12 for t and 0 for gt in $gt = at - 6^2 + 302.4$ yields $0 = a12 - 6^2 + 302.4$, or $36a = - 302.4$. Dividing both sides of this equation by 36 gives $a = - 8.4$. Substituting -8.4 for a in $gt = at - 6^2 + 302.4$ gives $gt = - 8.4t - 6^2 + 302.4$. Substituting 10 for t in gt gives $\mathrm{g}10 = - 8.410 - 6^2 + 302.4$, which is equivalent to $\mathrm{g}10 = - 8.44^2 + 302.4$, or $\mathrm{g}10 = 168$. Therefore, the estimated depth, to the nearest meter, of the seal 10 minutes after it entered the water was 168 meters.

Q4

D

D. $w = 94^{n - 1}$

Choice D is correct. Since w represents the nth term of the sequence and 9 is the first term of the sequence, the value of w is 9 when the value of n is 1. Since each term after the first is 4 times the preceding term, the value of w is 94 when the value of n is 2. Therefore, the value of w is 944, or 94^2, when the value of n is 3. More generally, the value of w is $94^{n - 1}$ for a given value of n. Therefore, the equation $w = 94^{n - 1}$ gives w in terms of n.

Choice A is incorrect. This equation describes a sequence for which the first term is 36, rather than 9, and each term after the first is 9, rather than 4, times the preceding term.

Choice B is incorrect. This equation describes a sequence for which the first term is 4, rather than 9, and each term after the first is 9, rather than 4, times the preceding term.

Choice C is incorrect. This equation describes a sequence for which the first term is 36, rather than 9.

Q5 C

C. 4

Choice C is correct. It's given that the function P models the population of the city t years after 2005. Since there are 12 months in a year, 18 months is equivalent to $\frac{18}{12}$ years. Therefore, the expression $\frac{18}{12}x$ can represent the number of years in x 18-month periods. Substituting $\frac{18}{12}x$ for t in the given equation yields $P\frac{18}{12}x = 2901.04^{\frac{4}{6}\frac{18}{12}x}$, which is equivalent to $P\frac{18}{12}x = 2901.04^{x}$. Therefore, for each 18-month period, the predicted population of the city is 1.04 times, or 104% of, the previous population. This means that the population is predicted to increase by 4% every 18 months.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. Each year, the predicted population of the city is 1.04 times the previous year's predicted population, which is not the same as an increase of 1.04%.

Choice D is incorrect and may result from conceptual or calculation errors.

Q6

A

A. $\frac{59}{5}$

Choice A is correct. The value of k for which $f k = 3k$ can be found by substituting k for x and $3k$ for $f x$ in the given equation, $f x = 59 - 2x$, which yields $3k = 59 - 2k$. For this equation to be true, either $-3k = 59 - 2k$ or $3k = 59 - 2k$. Adding $2k$ to both sides of the equation $-3k = 59 - 2k$ yields $-k = 59$. Dividing both sides of this equation by -1 yields $k = -59$. To check whether -59 is the value of k, substituting -59 for k in the equation $3k = 59 - 2k$ yields $3 - 59 = 59 - 2 - 59$, which is equivalent to $-177 = 177$, or $-177 = 177$, which isn't a true statement. Therefore, -59 isn't the value of k. Adding $2k$ to both sides of the equation $3k = 59 - 2k$ yields $5k = 59$. Dividing both sides of this equation by 5 yields $k = \frac{59}{5}$. To check whether $\frac{59}{5}$ is the value of k, substituting $\frac{59}{5}$ for k in the equation $3k = 59 - 2k$ yields $3\frac{59}{5} = 59 - 2\frac{59}{5}$, which is equivalent to $\frac{177}{5} = \frac{177}{5}$, or $\frac{177}{5} = \frac{177}{5}$, which is a true statement. Therefore, the value of k for which $f k = 3k$ is $\frac{59}{5}$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q7

14

The correct answer is 14. Let x represent the first integer and y represent the second integer. If the first integer is 5 greater than twice the second integer, then $x = 2y + 5$. It's given that the product of the two integers is 462; therefore $xy = 462$. Substituting $2y + 5$ for x in this equation yields $(2y + 5)(y) = 462$, which can be written as $2y^2 + 5y = 462$. Subtracting 462 from each side of this equation yields $2y^2 + 5y - 462 = 0$. The left-hand side of this equation can be factored by finding two values whose product is $2(-462)$, or -924, and whose sum is 5. The two values whose product is -924 and whose sum is 5 are 33 and -28. Thus, the equation $2y^2 + 5y - 462 = 0$ can be rewritten as $2y^2 - 28y + 33y - 462 = 0$, which is equivalent to $2y(y - 14) + 33(y - 14) = 0$, or $(2y + 33)(y - 14) = 0$. By the zero product property, it follows that $2y + 33 = 0$ or $y - 14 = 0$. Subtracting 33 from both sides of the equation $2y + 33 = 0$ yields $2y = -33$. Dividing both sides of this equation by 2 yields $y = -\frac{33}{2}$. Since y is a positive integer, the value of y isn't $-\frac{33}{2}$. Adding 14 to both sides of the equation $y - 14 = 0$ yields $y = 14$. Substituting 14 for y in the equation $xy = 462$ yields $x(14) = 462$. Dividing both sides of this equation by 14 yields $x = 33$. Therefore, the two integers are 14 and 33, so the smaller of the two integers is 14.

Q8

1728

The correct answer is $1{,}728$. The given function can be rewritten in the form $f x = a x - h^2 + k$, where a is a positive constant and the minimum value, k, of the function occurs when the value of x is h. By completing the square, $f x = x^2 - 48x + 2{,}304$ can be written as $f x = x^2 - 48x + \frac{48}{2}^2 + 2{,}304 - \frac{48}{2}^2$, or $f x = x - 24^2 + 1{,}728$. This equation is in the form $f x = a x - h^2 + k$, where $a = 1$, $h = 24$, and $k = 1{,}728$. Therefore, the minimum value of the given function is $1{,}728$.

Q9 A

A. $Vx = 11xx - 1$

Choice A is correct. The volume of a prism is equal to the area of its base times its height. It's given that the length of each house's base is x inches and that this length is 1 inch more than the width, in inches, of the house's base. It follows that the width, in inches, of the house's base is $x - 1$. The area of a rectangle is the product of its length and its width. Therefore, the area of the base of the house is $x(x - 1)$ square inches. It's given that the height of each house is 11 inches. Therefore, the function V that gives the volume of each house, in cubic inches, in terms of the length of the house's base x is $V(x) = x(x - 1)11$, or $V(x) = 11x(x - 1)$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q10 B

B. $fx = -3^x + 5$

Choice B is correct. The graph of $y = f(x) + 4$ is shown, and the graph passes through the point 0, 8. Substituting 0 for x and 8 for y in this equation yields $8 = f(0) + 4$. Subtracting 4 from both sides of this equation yields $4 = f(0)$. Therefore, when $x = 0$, the value of $f(x)$ is 4. Each of the given choices is in the form $fx = -3^x + k$, where k is a constant. Substituting 0 for x and 4 for $f(x)$ in this equation yields $4 = -3^0 + k$, or $4 = -1 + k$. Adding 1 to both sides of this equation yields $5 = k$. Substituting 5 for k in the equation $fx = -3^x + k$ yields $fx = -3^x + 5$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Notes

Accept equivalent forms (e.g. `0.25 ≡ 1/4`). For grid-in items, decimal and fraction answers are both valid.